

**DATA 3464: Fundamentals of Data Processing**

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# **Missing and weird data**

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# Topic overview

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- What to do with missing data
- Detecting and handling outliers

## Resources used:

- [Feature Engineering Chapter 8](#)
- Hands on Machine Learning with Scikit-Learn and Tensorflow/PyTorch, Chapter 2.  
Available at [MRU Library](#)
- [Scikit-learn user guide: Chapters 2 and 7](#)

# The problem

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- As you've seen, real-world data is messy
- Missing values are common, other values don't make sense
- We need to decide how to deal with these problems

*What examples have we seen so far?*

*Why might data be missing or weird\*?*

\*I'm using "weird" as an informal catch-all for unexpected or outlier values

# Missing data

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When data are missing in the *features* we have a few options:

0. Do nothing! Some algorithms (e.g. [Decisions Trees](#)) can handle missing values
1. Remove *features* with missing values
2. Remove *samples* with missing values
3. Invent a new value to represent "missingness"
4. **Impute** a value based on other data

*Most important: understand why data are missing (more EDA!)*

# Option 1: removing features

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- If a feature has:
  - A high proportion of missing values, *and*
  - Little apparent relationship to the target *or*
  - its information is redundant with other features
- It may be reasonable to remove it entirely. You can drop it, e.g.:

```
df.drop(columns=['feature_name'], inplace=True)
```

or (probably more reliable) just not select it when building your pipeline

# Option 2: removing samples

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- If only a small number of samples have missing values, you can drop them **from the training data**:

```
train.dropna(subset=[["features", "we", "care", "about"]], inplace=True)
```

- Good idea if the same samples have missing values from multiple features
- Still useful to explore *why* data are missing

*What should we do for inference?*

# Option 3: invent a new value

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- Categorical features: add a new category for "missing"
- Add a new binary feature indicating whether the value was missing
- I have seen advice to use extreme values for numerical features, like the -1 income in the OKCupid dataset, but I'm not convinced this is a good idea

*Case study: where missingness is informative*

# Option 4: impute missing values

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- Fill in the missing values with an "educated guess"
- Replace missing value with:
  - constant
  - mean, median, or mode ( `most_frequent` )
- Use other features to infer missing value:
  - K-nearest neighbours
  - simple models to predict missing values
- Can be combined with option 3 to indicate missing features
- How much to impute? [Feat.Engineering](#) suggests no more than 20%



# Choosing an imputation strategy

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| Strategy              | When to use                               |
|-----------------------|-------------------------------------------|
| Constant              | When there is a reasonable default value  |
| Mean                  | Numeric features with normal distribution |
| Median                | Numeric features with extreme outliers    |
| KNN                   | Relationship with other features          |
| Missingness indicator | If missingness seems informative          |

# Outliers

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*An outlier is an observation which deviates so much from the other observations as to arouse suspicions that it was generated by a different mechanism. -- D. M. Hawkins*

- There are many entire books dedicated to outlier detection
- Useful for anomaly detection, e.g:
  - fraud detection
  - failure prediction
  - **Many other applications**
- Our focus is on dealing with outliers in preprocessing

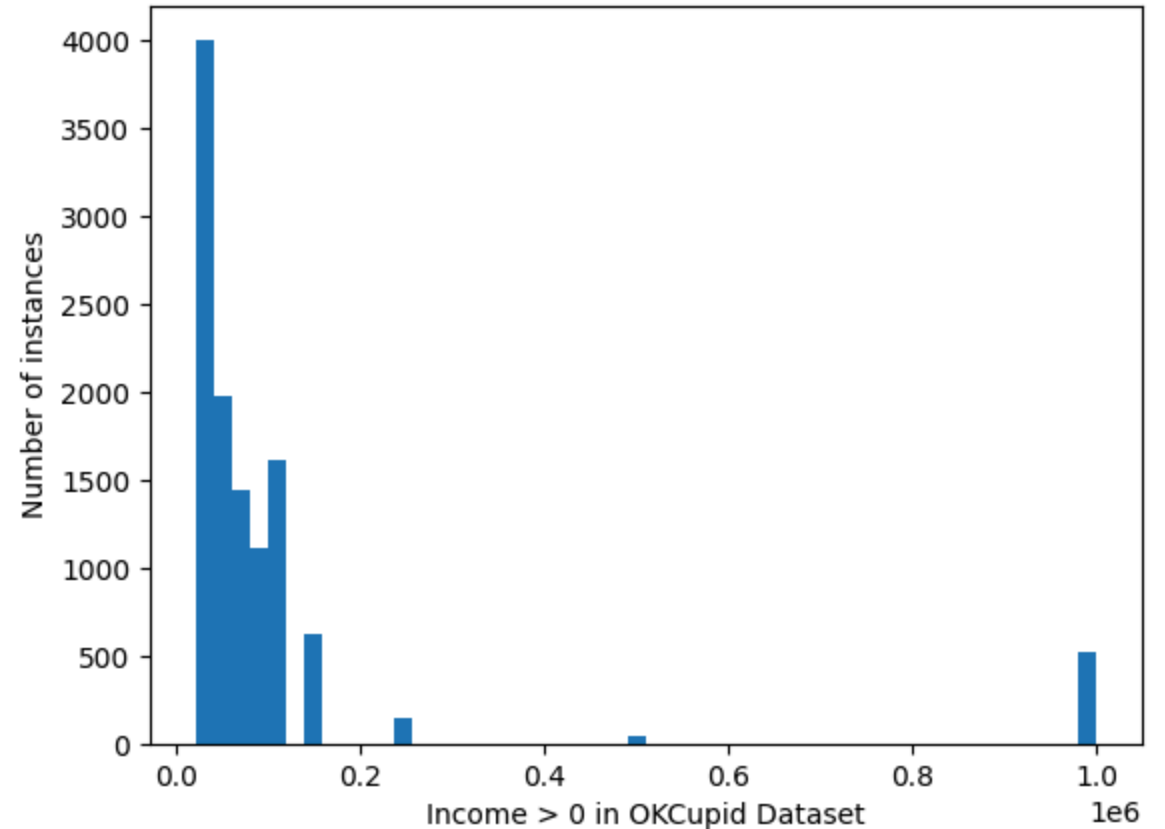
# Where we left off on February 5

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# Detecting outliers

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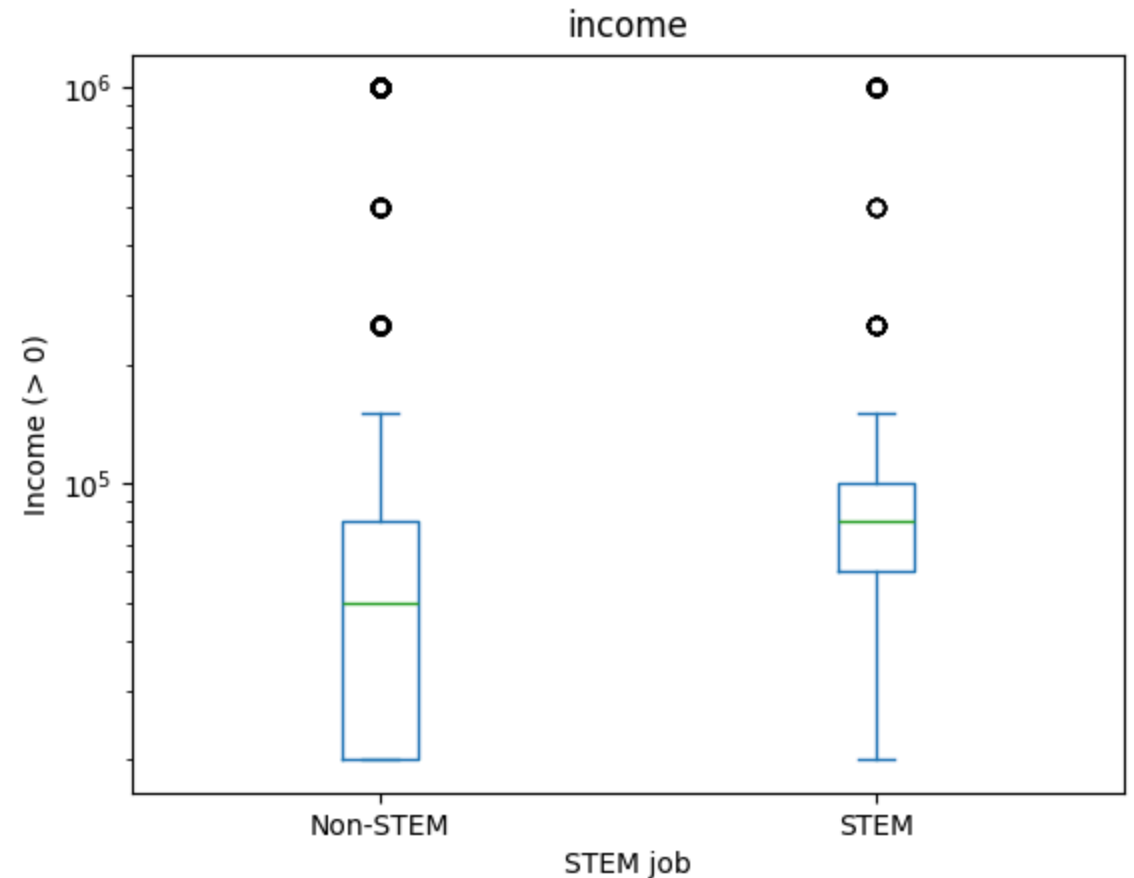
- Visually as part of EDA
- Statistically, e.g.  $> 3\sigma$
- Algorithmically, e.g. [Isolation Forests](#)
- As usual, context and domain knowledge are essential



# Context matters!

- Look at the relationship between outliers and target (training dataset, of course)

*What do the dots on box plots represent?*



# What to do about outliers?

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1. Data transformations
2. Drop the samples
3. Encode them somehow
4. Leave them alone

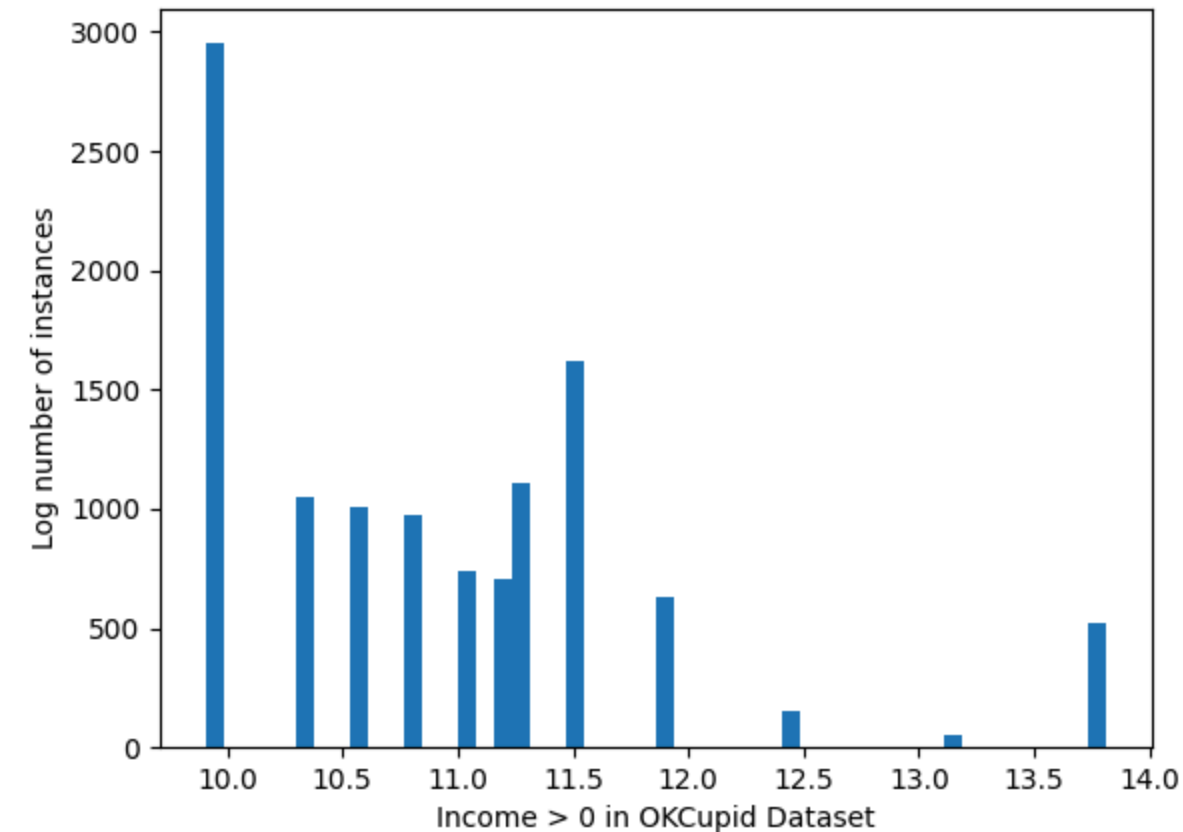
As usual, very data- and model-dependent. Tree based methods are particularly impacted by outliers!

*Any other ideas?*

# Nonlinear transformations

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- Transforming the data does not actually remove the outlier
- Can help make the relationship less extreme



# Dropping the samples

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- In general, not a thing I love to do
- If you drop a sample from training, you need to decide what to do at inference
- My opinion: Only do it if you're confident it's an error in the dataset

*Can you think of an example?*

- What can you do at inference time when outliers are encountered?



# Encoding outliers

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A few other options that might fall into "encoding":

- Just like with missing values, binary column indicating outlier/inlier
- Remove the values and convert them to missing
- Bin or impose a cap (floor/ceiling) on the value
- Replace the numeric value with a rank or quantile bucket
- Probably other things!

# Leave them alone

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If your outliers are:

- Real values (not data entry or other errors)
- Representative of things that might happen during inference

Then you probably want to keep them!

*Consider using a [RobustScalar](#) to standardize if you have lots of outliers*

# Missing values and outliers in the target

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- If you have missing values in the target, you probably want to avoid imputing
- This is a good case for dropping samples from **training**!
- Outliers are trickier -- again, check if they're real or mistakes
- There may be a case for transforming the target

# Coming up next

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- Interactions between variables
- Feature selection
- Midterm stuff
- Reading week!!!!